

Problem Set 3: Numerical Integrals and Derivatives

Readings: Newman Ch. 5

Due: Wednesday, September 30th 11:59 PM

Instructions: Work through each problem, following the instructions for the requirements of the code and their outputs. Some starting code is provided for each problem on the course website. Problem sets should be submitted in one batch, with each problem uploaded as it's own python script (.py file) on the HuskyCT assignment page with the filename: `ps[PROBLEM SET NUMBER]_[QUESTION NUMBER]_[LASTNAME].py`.

Parameters:

- (1) Submitted solutions are restricted to imports of external functions from standard built-in Python libraries (i.e. `math`, `time`, `os`, `sys`, etc.). In addition, imports from `matplotlib.pyplot` and `numpy` are permitted for basic array operations.
 - (2) All plots should have both axes clearly labeled and include a title. Plots should include either legends or annotations for differentiating plot elements, when applicable.
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1. HEAT CAPACITY OF A SOLID

Debye's theory of solids gives the heat capacity of a solid at temperature T to be

$$C_V = 9Vnk_B \left(\frac{T}{\theta_D} \right)^3 \int_0^{\theta_D/T} \frac{x^4 e^x}{(e^x - 1)^2} dx,$$

where V is the volume of the solid, n is the number density of atoms, k_B is Boltzmann's constant, and θ_D is the so-called *Debye temperature*, a property of solids that depends on their density and speed of sound.

Write a program to compute the heat capacity for a sample consisting of 1000 cubic centimeters of solid aluminum, which has an atomic number density of $n = 6.022 \times 10^{28} \text{ m}^{-3}$ and a Debye temperature of $\theta_D = 428 \text{ K}$.

The program should:

- a. Include a Python function `cv(T)` that calculates C_V for a given value of the temperature in Joules per Kelvin using Gaussian quadrature to evaluate the integral, with $N = 50$ sample points.
- b. Use your function to print the value of C_V at room temperature, $T = 295 \text{ K}$.
- c. Make a plot of the heat capacity as a function of temperature from $T = 5 \text{ K}$ to $T = 500 \text{ K}$.
- d. Save the figure as `Debye_[LASTNAME].png`

Hint: you can use the gaussian quadrature module from our in-class exercises.

2. PERIOD OF AN ANHARMONIC OSCILLATOR

The simple harmonic oscillator (SHO) crops up in many parts of physics. Its behavior can be studied readily using analytic methods and it has the important property that its period of oscillation is a constant, independent of its amplitude, making it useful, for instance, for keeping time. Frequently in physics, however, we also come across anharmonic oscillators, whose period varies with amplitude and whose behavior cannot usually be calculated analytically.

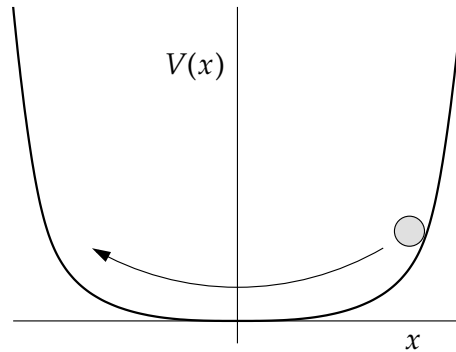


Figure 1: A general classical oscillator can be thought of as a particle in a concave potential well. When disturbed, the particle will rock back and forth in the well.

The simple harmonic oscillator corresponds to a quadratic potential $V(x) \propto x^2$, whereas any other form gives an anharmonic oscillator. (Thus there are many different kinds of anharmonic oscillator, depending on the exact form of the potential.) One way to calculate the motion of an oscillator is to write down the equation for the conservation of energy in the system. If the particle has mass m and position x , then the total energy is equal to the sum of the kinetic and potential energies is thus:

$$E = \frac{1}{2}m\left(\frac{dx}{dt}\right)^2 + V(x).$$

Since the energy must be constant over time, this equation is effectively a (nonlinear) differential equation linking x and t .

Let us assume that the potential $V(x)$ is symmetric about $x = 0$ and the anharmonic oscillator is initially at rest with some initial amplitude a . That is, at $t = 0$, we release it from rest at position $x = a$ and it swings back towards the origin. Then at $t = 0$ we have $dx/dt = 0$ and the equation above reads $E = V(a)$, which gives us the total energy of the particle in terms of the amplitude. When the particle reaches the origin for the first time, it has gone through one quarter of a period of the oscillator. By rearranging the equation above for dx/dt and then integrating with respect to t from 0 to $\frac{1}{4}P$, it can be shown that the period P is given by:

$$P = \sqrt{8m} \int_0^a \frac{dx}{\sqrt{V(a) - V(x)}}.$$

Write a program to calculate the period of an anharmonic oscillator with the potential $V(x) = x^4$ and $m = 1$. The program should:

- Include a function that calculates the period P of the oscillator for given amplitude a the using Gaussian quadrature method with $N = 20$ points.
- Print the results of your function for an amplitude of $a = 1$.
- Make a plot of the period P for amplitudes ranging from $a = 0$ to $a = 2$.
- Save the figure as `anharmonic_[LASTNAME].png`.

3. THE STEFAN–BOLTZMANN CONSTANT

The Planck theory of thermal radiation tells us that in the (angular) frequency interval ω to $\omega + d\omega$, a black body will radiate an amount of thermal energy per area per second equal to $I(\omega)d\omega$, where

$$I(\omega) = \frac{\hbar}{4\pi^2 c^2} \frac{\omega^3}{(e^{\hbar\omega/k_B T} - 1)}.$$

Here $\hbar$ is Planck's constant over 2π , c is the speed of light, and k_B is Boltzmann's constant. Integrating over ω , it can be shown that the total flux: the energy per unit area per time radiated by a black body is

$$F = \frac{k_B^4 T^4}{4\pi^2 c^2 \hbar^3} \int_0^\infty \frac{x^3}{e^x - 1} dx.$$

An expression consistent with the well-known Stefan-Boltzmann law for the total flux from a blackbody: $F = \sigma T^4$, where σ is the Stefan–Boltzmann constant.

Write a program to evaluate the integral for the total flux, F . The program should:

- Include a function to calculate the value of the integral as a function of the temperature T using an integration method of your choice.
- Calculate and print the value of the integral to calculate the total flux emitted by the Sun ($T \sim 6000$ K).
- Calculate and print the error in your numerical integral using an error determination method of your choice.
- Use your value for the integral above to compute and print out a value for the Stefan–Boltzmann constant σ (in SI units) to three significant figures.

4. ELECTRIC FIELD OF A CHARGE DISTRIBUTION

Suppose we have a distribution of charges and we want to calculate the resulting electric field. One way to do this is to first calculate the electric potential ϕ and then take its gradient. For a point charge q at the origin, the electric potential at a distance r from the origin is

$$\phi = q/4\pi\epsilon_0 r$$

and the corresponding electric field is

$$\vec{E} = -\nabla\phi$$

Suppose you have two charges, of $q = \pm 1$ C, placed some distance d apart from one another.

Write a program to compute the electric potential and the resulting electric field across an $L \times L$ square surface in the plane of the two charges. Your program should:

- Include a function to calculate the resulting electric potential for two charges spaced d apart on a grid in $x - y$ from $-L/2 \leq x \leq L/2$, $-L/2 \leq y \leq L/2$ for $L = 1$ m, and grid points spaced 1 cm apart.
- Use your function to calculate the electric field for a charge separation of $d = 10$ cm and make a density plot figure of the potential field on the surface plane. Indicate the locations of the two charges by plotting them as

points or annotating. This plot should have a colorbar. You will want to limit the magnitude of the colormap using `vmin` and `vmax` in order to be able to show the variation in the field.

- c. Save this figure as `epot_[LASTNAME].png`.
- d. Calculate the partial derivatives of the potential with respect to x and y using a central differences scheme to find $\vec{E}(x, y)$, the electric field in the xy plane.
- e. Make a figure visualizing the resulting electric field. The figure should show both the magnitude and direction of the electric field. You can either do this by making a two panel density plot for each quantity or by overplotting the direction of the field as arrows on top of a density plot of the magnitude using `matplotlib`'s `quiver` function or something similar.
- f. Save this figure as `efield_[LASTNAME].png`.

Hint: you'll need to use an alternative scheme to find the derivatives at the boundary points